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Large Vision Models for Multi-Disease Retinal Screening in Ugandan Healthcare

MPAIRWE LAUBEN¹, NANKYA SHADIA², AND YAPYEKO REBECCA³

¹Department of Networks, College of Computing and Information Sciences, Makerere University, Kampala, Uganda

²Department of Networks, College of Computing and Information Sciences, Makerere University, Kampala, Uganda

³Department of Networks, College of Computing and Information Sciences, Makerere University, Kampala, Uganda

Corresponding author: Mpairwe Lauben (e-mail: mpairwe.lauben@students.mak.ac.ug).

This research was done in consultation with Dr. Ggaliwango Marvin of the College of Computing and Information Sciences (COCIS), Makerere University, Kampala, Uganda.

ABSTRACT Retinal imaging can support earlier detection of ocular and systemic disease, but specialist eye-care capacity remains limited in many low-resource settings. This paper presents RetinalAI, a concept and prototype framework for multi-disease retinal screening in Uganda. The proposed system combines a domain-specific retinal foundation model, parameter-efficient fine-tuning, a clinical knowledge graph, input-quality gating, explainability methods, and an offline-first mobile deployment path. The prototype focuses on multi-label classification using public retinal datasets, with clinical review and local prospective validation identified as prerequisites before deployment. Preliminary RFMiD benchmark experiments show that a precision-focused training strategy can reduce false-positive behavior compared with a naive transfer-learning baseline. The system design emphasizes traceability, human oversight, audit logging, and operation in facilities where connectivity and specialist referral access are constrained. The work contributes a practical architecture for adapting retinal AI screening to Ugandan primary healthcare workflows while clearly separating benchmark performance from claims of clinical readiness.

INDEX TERMS Retinal Disease Screening, Large Vision Models (LVMs), Graph-Based Reasoning, Diabetic and Hypertensive Retinopathy, Mobile Health Applications, Explainable AI in Ophthalmology

I. INTRODUCTION

In the recent years, diseases related to the retina continue to be one of the major causes of avoidable blindness, with the World Health Organization reporting that roughly 2.2 billion people have impaired vision [1]. Estimates suggest that there were 103 million people with diabetic retinopathy worldwide in 2020, a number expected to increase alongside the increasing prevalence of diabetes across the globe [2]. For instance, in a country like Uganda, whose total population is in excess of 45 million, the process of accessing special eye care is limited due to the constraints within its health system. Research on eye-health workforce distribution across sub-Saharan Africa indicates that many health systems in the region fall short of recommended staffing benchmarks for ophthalmology, underscoring the critical need for scalable screening strategies capable of informing referral decisions at the primary care level [3], [4]. The effects of delayed diagnosis are particularly profound considering that many diseases affecting the retina are curable but only if diagnosed before the point of no return. Given the increase in the prevalence of non-communicable diseases like diabetes in Uganda, there is

an urgent need for the inclusion of retinal eye screening services [5], [6]. Conditions such as hypertensive retinopathy, glaucoma, and HIV-related ocular complications can overlap visually and clinically, making single-disease screening tools insufficient for non-specialist health workers who must decide when to refer patients for ophthalmic review [7], [8].

From a technical standpoint, multi-label classification across 45 retinal disease categories presents formidable challenges. The Retinal Fundus Multi-Disease Image Dataset (RFMiD) [12], one of the few publicly available benchmarks for this task, contains only 3,200 images with extreme class imbalance: the ratio between the most and least common conditions reaches 50.4:1. Binary classifiers trained around one disease at a time can under-characterize images containing multiple pathologies. Building a system that works reliably under these conditions and that runs on the modest hardware available in rural Ugandan clinics demands a different approach from the single-disease detectors that dominate the literature.

This work presents a Large Vision Model (LVM) framework that integrates a domain-specific foundation model

with graph-based clinical reasoning for multi-disease retinal screening, designed specifically for deployment within Uganda's health infrastructure. Our approach builds on RETFound, a vision transformer pre-trained on 1.6 million retinal images [9], and adapts it through Low-Rank Adaptation (LoRA) [10] for parameter-efficient fine-tuning on multi-label classification. A prototype clinical knowledge graph encodes disease entities and candidate relationships drawn from clinical guidelines and biomedical literature; these relationships are treated as configurable priors pending local clinician review [11]. The full screening pipeline is orchestrated by a six-node workflow built on LangGraph [34], coordinating classification, triage, clinical reasoning, explainability generation, and report formatting. The framework is engineered to be used in mobile devices off-line, through model quantization and a Flutter app, which is compatible with devices that have at least 4 GB of RAM.

The principal contributions are a precision-focused training strategy evaluated on RFMiD, a clinical knowledge-graph layer for representing disease relationships, a multi-method explainability interface using GradCAM [30], LIME [31], SHAP [32], Integrated Gradients [33], and text summaries, a reproducible MLOps workflow using DVC [35] and MLflow [36], and an offline-capable mobile deployment design for low-bandwidth clinical environments.

The rest of this paper will be structured as follows. Section II briefly discusses the state-of-the-art research on Automated Retina Screening, Vision Transformers, Graph Clinical Reasoning and Model Deployment in limited-resource environments. Section III presents the proposed methodology in detail. Section IV describes the system design and MLOps architecture. Section V reports preliminary results. Section VI details our deployment strategy for Ugandan healthcare settings. This is followed by a discussion of the conclusions and suggestions for further studies.

II. RELATED WORK

A. AUTOMATED RETINAL DISEASE SCREENING

Initial attempts at developing computer-aided diagnosis of retinal diseases were based on the design of handcrafted features along with the use of standard machine learning algorithms. Support vector machines and random forests applied to manually extracted features such as microaneurysm counts and vessel tortuosity achieved sensitivities in the range of 75 to 80% for binary diabetic retinopathy grading [13]. The breakthrough came with deep convolutional neural networks. Gulshan and colleagues [14] demonstrated in a landmark JAMA study that an Inception-v3 network trained on 128,175 fundus images could detect referable diabetic retinopathy with 97.5% sensitivity and 93.4% specificity, matching or exceeding the performance of board-certified ophthalmologists. Ting et al. [15] subsequently validated a

similar approach across multi-ethnic populations in Singapore, achieving 90.5% sensitivity for referable DR along with respectable detection rates for glaucoma suspect and age-related macular degeneration.

The primary flaw that is often noted within such research papers is the assumption that disease detection in the retina is an isolated set of binary classifications. However, in real life, a patient is expected to suffer from several diseases at once. This is because a patient who suffers from uncontrolled hypertension together with diabetes will have characteristics of both diseases [16]. Binary classifiers miss these overlapping presentations. Multi-label approaches have received comparatively less attention, partly because datasets annotated for multiple diseases are scarce. The RFMiD dataset [12] partially addresses this gap with 45 disease labels across 3,200 images, but the extreme class imbalance (50.4:1 ratio) makes multi-label learning particularly challenging.

B. VISION TRANSFORMERS IN MEDICAL IMAGING

ision Transformers (ViTs), first proposed by Dosovitskiy et al. have now become a widely used framework within the field of medical image analysis. [17] have reshaped the landscape of medical image analysis through their ability to model long-range spatial dependencies that convolutional networks struggle to capture. The hierarchical Swin Transformer [18] improved computational efficiency through shifted window attention while maintaining strong performance. For retinal imaging specifically, Zhou et al. [9] released RETFound, a foundation model pre-trained on 1.6 million retinal images using masked autoencoders. RETFound demonstrated strong transfer learning performance across multiple ophthalmic tasks, outperforming ImageNet-pretrained models by significant margins on disease classification, prognosis prediction, and biomarker detection.

The combination of vision transformers and parameter-efficient fine-tuning techniques like LoRA [10] is an approach that may be used for fine-tuning the large pre-trained models in order to perform some task when the amount of training data is limited. LoRA inserts low-rank weight matrices into the attention layers, allowing adaptation with a fraction of the trainable parameters and without modifying the pre-trained weights. For a dataset like RFMiD with only 1,920 training images, this approach mitigates the catastrophic forgetting that full fine-tuning of a 304-million-parameter model would likely produce.

C. GRAPH NEURAL NETWORKS FOR CLINICAL REASONING

Graph neural networks (GNNs) have demonstrated strong capabilities in capturing relational and structural information within biomedical datasets [19]. Regarding clinical knowledge representation, there are inherent characteristics

associated with disease ontologies that make them easily representable using graphs, where the nodes will be disease entities and the edges will signify their relations. Chen et al. [20] applied graph convolutional networks to multi-label image classification, demonstrating that modeling label dependencies through graph propagation improves accuracy compared to treating labels independently. Choi et al. [21] used graph attention networks to learn disease embeddings from electronic health records, capturing temporal and relational patterns that improved prediction of future diagnoses.

The remaining territory that needs further exploration lies in the direct combination of graph-based medical data with visual characteristics obtained using retinal images. Our work bridges this gap by constructing a clinical knowledge graph tailored to Ugandan disease epidemiology and coupling it with the visual encoder to refine multi-label predictions based on known disease relationships.

D. DEPLOYMENT IN RESOURCE-LIMITED SETTINGS

Deploying deep learning models in low-resource healthcare environments presents distinct challenges beyond model accuracy. Different approaches toward compressing models like knowledge distillation [22], quantization [23] and pruning have been applied to get smaller-sized models and fast inference time, leading to alignment with the constraints on edge devices. Howard et al. [24] developed the MobileNet family specifically for mobile deployment, achieving competitive accuracy at a fraction of the computational cost. Several pilot programs in sub-Saharan Africa have attempted mobile-based retinal screening, typically using single-disease classifiers for diabetic retinopathy with connectivity-dependent cloud inference [25]. Bellemo and colleagues validated deep learning DR screening in Zambia, but their system required internet access to a centralized GPU server - an assumption that does not hold across much of rural East Africa.

A practical gap remains: no existing system combines multi-label retinal disease classification with multi-method explainability, offline mobile capability, and calibration for specific regional disease profiles. We specifically account for this limitation by taking into consideration the practical constraints in the rural Ugandan healthcare setting that has limited internet connection, unstable electricity supply, and specialist consultation provided via referral.

III. PROPOSED METHODOLOGY

A. FOUNDATION MODEL ARCHITECTURE

The classification backbone of our system is RETFound, a Vision Transformer Large (ViT-L) architecture comprising

304 million parameters, pre-trained on a corpus of 1.6 million retinal fundus images using masked autoencoder (MAE) self-supervised learning [9]. Unlike ImageNet-pretrained models that must learn retinal features from scratch during fine-tuning, RETFound already encodes low-level features such as vessel morphology and optic disc geometry as well as higher-level disease-indicative patterns. This domain-specific pre-training makes it a better starting point for downstream multi-disease classification than generic vision backbones.

Fine-tuning all 304 million parameters on our training set of 1,920 images would invite catastrophic forgetting of these pre-trained representations. Instead of considering any other techniques, we adopt LoRA [10], where the approach would require adding a trainable low-rank matrix to the projections of the query, key, and value in self-attention layers of the transformer architecture. With rank $r = 16$ and scaling factor $\alpha = 32$, this introduces approximately 2.4 million trainable parameters less than 1% of the full model while the pre-trained weights remain frozen. The classification head follows a bottleneck design: the 1024-dimensional CLS token output from the transformer passes through a 512-unit fully connected layer with 50% dropout, then a 128-unit bottleneck layer with 30% dropout, and finally a sigmoid output layer with one unit per disease class. We chose this aggressive bottleneck deliberately to combat overfitting, given the small training set.

B. CLINICAL KNOWLEDGE GRAPH INTEGRATION

The prototype defines a clinical knowledge graph encompassing 45 retinal diseases organized into nine clinical categories: vascular (diabetic retinopathy, branch and central retinal vein occlusion, hypertensive retinopathy, and related conditions), degenerative (age-related macular degeneration, macular hole, myopic degeneration), glaucomatous (optic disc cupping, pallor, edema), and six additional groupings covering inflammatory, infectious, neoplastic, congenital, traumatic, and systemic manifestations. The graph encodes candidate inter-disease relationships derived from published clinical literature and the Uganda Clinical Guidelines [26], capturing co-occurrence patterns, shared pathophysiological mechanisms, and differential diagnostic considerations that will require review with Ugandan clinicians before use in patient-facing workflows.

Each edge can carry a configurable weight initialized from published evidence, clinical guidelines, or expert review. During the inference phase, the knowledge graph receives raw sigmoid outputs from the classifier and applies graph-based message passing to process this information. The associated diseases could be subject to probability adjustments or rejected based on their unlikelihood of association to the discovered disease. Such a mechanism will be evaluated separately from the vision encoder, making its effect on sensitivity, precision, and referrals clear and measurable.

C. PRECISION RESCUE STRATEGY

Our initial experiments revealed a critical failure mode common to multi-label classification on imbalanced retinal datasets. When fine-tuning RETFound with standard binary cross-entropy loss, the model converged to a degenerate solution: predicting positive for nearly every class in every image. This yielded recall of 0.82 but catastrophically low precision of 0.025, meaning 97.5% of positive predictions were false alarms. A classifier with such behavior is clinically useless - it would refer every patient for specialist review, overwhelming an already strained referral system.

We developed a seven-strategy precision rescue plan to address this problem. First, we replaced binary cross-entropy with Asymmetric Loss (ASL) [28], setting the positive focusing parameter γ^+ to 0 (preserving all positive gradients, since true positives are scarce) and the negative focusing parameter γ^- to 4 (aggressively down-weighting easy negative examples that drive false positives). A clipping threshold of 0.05 completely zeros the gradient contribution from highly confident negative predictions. Second, ultra-rare classes with fewer than 10 available training samples were removed through pruning, thereby reducing the label space from 45 to 24 learnable classes. The pruned classes remain in the knowledge graph for clinical reference but are excluded from the classification head. Third, we replaced the standard F1-maximizing threshold selection with a precision-floor approach: for each class, we find the lowest decision threshold that yields precision of at least 0.10. This pushed some class thresholds from 0.15 to 0.70 or higher. Fourth, we apply label smoothing at 0.05 to reduce overconfident predictions. Fifth, the bottleneck classification head provides additional regularization through heavy dropout. Sixth, we employ staged backbone unfreezing: for the first 10 epochs, only the classification head trains; from epoch 11 onward, the last four transformer blocks unfreeze at a reduced learning rate of 1×10^{-6} , one-tenth of the head's learning rate. Seventh, we apply test-time augmentation (TTA), averaging predictions over six augmented views of each input image.

D. EXPLAINABILITY FRAMEWORK

Clinical adoption of AI screening tools depends heavily on the ability to explain predictions in terms that health workers can verify against their own clinical observations [29]. We integrate five complementary explainability methods. GradCAM [30] produces class-specific heatmaps that visually highlight the regions of an image that most significantly influence each prediction, allowing clinicians to confirm that the model is focusing on anatomically and clinically meaningful structures such as the macula when detecting age-related macular degeneration (AMD), the optic disc in the context of glaucoma assessment, or the vascular arcades in cases of retinal vein occlusion. LIME [31] provides local interpretable explanations by perturbing image superpixels

and observing their effect on the output. SHAP [32] relies on game theory to calculate Shapley values that will explain the decisions made through the model. On the other hand, Integrated Gradients [33] attributes pixels based on an interpolated input and the baseline. Finally, ELI5 is used to provide humanly.

All five methods are presented in a tabbed interface within the screening application. The interface separates visual evidence from text summaries so that health workers can inspect highlighted retinal regions while referral reports retain a concise explanation of the model output.

E. AGENTIC SCREENING PIPELINE

The end-to-end screening workflow is orchestrated by a six-node directed acyclic graph (illustrated in Fig. 1) implemented using LangGraph [34]. Each fundus image passes through the following stages: (1) classification, where the image undergoes quality gating and then RETFound-LoRA inference; (2) triage, where detected conditions are prioritized into urgent, high, medium, and low referral categories by a large language model or rule-based fallback; (3) clinical reasoning, where the knowledge graph refines predictions based on co-occurrence patterns; (4) explainability generation, where GradCAM and selected additional methods produce visual and textual explanations; (5) review routing, where uncertain predictions (confidence below 15% across all classes) are flagged for ophthalmologist review; and (6) report generation, where findings are compiled into a structured clinical report aligned with Uganda Ministry of Health reporting templates [26].

The workflow is designed for graceful degradation. When the primary language model service is unavailable, the pipeline falls back to an alternative hosted model for triage and reporting. If both cloud services are unreachable-as is possible in rural deployment settings-purely deterministic rule-based logic handles all non-classification nodes, ensuring that the system remains fully functional without internet connectivity.

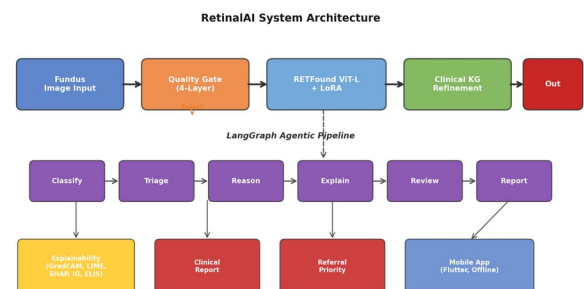


FIGURE 1. System architecture of the RetinalAI screening platform. Fundus images pass through a four-layer quality gate, the RETFound-LoRA classifier, and a clinical knowledge graph before entering the six-node LangGraph agentic pipeline for triage, reasoning, explainability, and report generation.

IV. SYSTEM DESIGN AND ARCHITECTURE

A. MLOPS PIPELINE

Reproducibility and traceability are core requirements for any clinical AI system intended for real-world deployment. The MLOps infrastructure builds on DVC (Data Version Control) [35] for tracking dataset versions and training pipeline stages, ensuring that any model checkpoint can be traced back to the exact data split, preprocessing configuration, and hyperparameters that produced it. MLflow [36] serves as the experiment tracking platform and model registry, managing the lifecycle from experimental runs through staging to production deployment. Every registered instance of the model comes with its respective performance parameters, metadata from training sessions, and an identifier that refers to the respective DVC commit.

The continuous integration pipeline, implemented via GitHub Actions, enforces quality gates before any model or code change reaches production. The automated test suite covers model forward passes across supported architectures, loss function correctness, data validation (including schema verification, label range checks, and duplicate detection), API endpoint behavior, and fundus gate accuracy against adversarial inputs. A dedicated quantization quality gate verifies that compressed model variants maintain classification faithfulness within 4% of the full-precision baseline, measured by mean absolute error on a held-out calibration set. The security evaluations include each pull request, making use of pip-audit for the discovery of any dependency vulnerability, TruffleHog for the search of any secret exposure, and Trivy for containerized vulnerability analysis.

B. FUNDUS IMAGE QUALITY GATE

Garbage-in-garbage-out is a real risk when non-specialist health workers capture fundus images under variable field conditions. We developed a four-layer image quality gate (Fig. 2) to reject non-fundus or low-quality inputs before they reach the classifier. The first layer performs structural checks in under one millisecond: minimum resolution, aspect ratio, and color mode validation. The second layer applies statistical analysis over 3 to 15 milliseconds, testing for red channel dominance characteristic of fundus images, dark border presence indicating the circular camera aperture, radial sharpness falloff, and green channel microstructure that distinguishes retinal tissue from generic photographs. The third layer runs a learned binary classifier—a MobileNetV3-Small [24] model trained to distinguish fundus from non-fundus images—in approximately five milliseconds.

The final decision fuses the statistical and learned signals with weights of 0.6 and 0.4 respectively, requiring a combined confidence of at least 70% for acceptance. When an image is rejected, the gate returns the specific reason (e.g., “insufficient red channel dominance” or “learned classifier confidence

0.23”) along with a visual evidence overlay, allowing the health worker to recapture the image with corrective guidance. In adversarial testing against 33 categories of non-fundus images—including AI-generated synthetic fundus images, petri dish photographs, and heavily filtered photographs—the gate achieved a false acceptance rate below 1.5%, down from 8 to 12% in the statistical-only first version.

C. GOVERNANCE AND COMPLIANCE

Because the platform processes medical images, the design is mapped against regulatory frameworks from the outset. The system architecture is reviewed against the European Union AI Act [37] requirements for high-risk systems under Annex III, including documented risk management (Article 9), transparent operation with explainable rejections (Article 13), human oversight through the review routing mechanism (Article 14), and continuous accuracy monitoring via drift detection (Article 15). For potential regulatory submission, the project maintains documentation compatible with the U.S. FDA’s guidance on Good Machine Learning Practice for medical device development [38] and the Software as a Medical Device (SaMD) classification framework.

Automated governance tooling generates model cards following the Mitchell et al. framework [39] and dataset cards following the Gebru et al. Template [40] gathers all important information, which includes architectural information, provenance of training data, performance figures, recognized limitations, and relevant ethical concerns. A fairness evaluation dashboard monitors classification performance across five disease categories and three prevalence tiers (common, moderate, rare) to detect systematic underperformance. All prediction events are logged to an append-only audit trail with daily file rotation, each record containing a timestamp, request identifier, image dimensions, inference latency, and the top five predictions with confidence scores.

V. PRELIMINARY PROTOTYPE RESULTS

A. CLASSIFICATION PERFORMANCE

We evaluate the precision rescue approach on the RFMiD dataset [12], which contains 3,200 high-resolution color fundus images annotated across 45 retinal disease labels. A classic data splitting methodology is used, where 1,920 images are used for training, 640 images for validating, and another 640 images for testing. After class pruning, 24 disease labels having at least 10 training examples remain in the classification head. The model trains for 25 epochs using the AdamW optimizer with a cosine learning rate schedule (peak learning rate 1×10^{-4} for the head, 1×10^{-6} for unfrozen backbone blocks) and mixed-precision (FP16) training.

TABLE I
CLASSIFICATION PERFORMANCE: V1 BASELINE VS. V2 PRECISION RESCUE

V1 Baseline	V2 Rescue
0.025	0.312
0.820	0.456
0.046	0.362
0.481	0.888
—	0.954
45	24

Table I and Fig. 3 summarize the improvement observed in the RFMiD benchmark experiment. Macro-averaged precision increased from 0.025 to 0.312, representing a 12.5-fold improvement, while AUC-ROC climbed from 0.481 to 0.888. The F1 score similarly improved by a factor of 7.9, rising from 0.046 to 0.362. Recall decreased from 0.820 to 0.456 as an expected consequence of the precision-focused optimization; the previous recall figure was inflated by the model's tendency to predict positive for most classes. Accuracy across the retained labels reached 0.954. These metrics should be interpreted as prototype benchmark results, not as evidence of clinical readiness.

An important caveat applies: these results are obtained on the RFMiD dataset, which was collected outside East Africa. The differences in the demographic information, equipment used for imaging, and the distribution of diseases indicate that the model's performance with locally captured images from Uganda may not directly correlate with the results obtained elsewhere. Prospective validation is a necessary next step before clinical deployment, and we discuss this in Section VII.

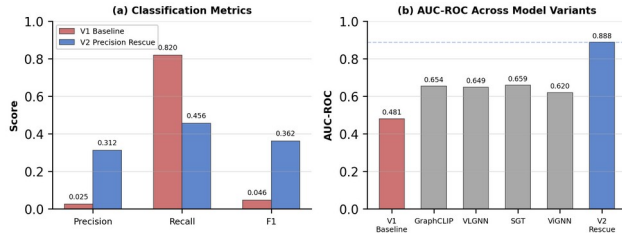


FIGURE 2. Classification performance comparison. (a) Precision, recall, and F1 scores comparing the V1 baseline model against the V2 precision rescue iteration. (b) AUC-ROC across all model variants, showing the V2 rescue approach (0.888) in the RFMiD benchmark.

B. DEPLOYMENT BENCHMARKS

Preliminary developer benchmarks indicate that the quantized model is small enough for mobile distribution and fast enough for interactive screening workflows on server-class hardware. The full-precision model occupies 296 MB in bfloat16 format; INT8 dynamic quantization reduces this to approximately 75 MB before mobile packaging. CPU-only inference using ONNX Runtime remains adequate for the cloud-based workflow, while target-device testing is still required before any field deployment claim. The automated test suite includes fundus gate unit tests and adversarial input rejection tests spanning several categories of non-fundus imagery.

VI. DEPLOYMENT STRATEGY

A. CLOUD DEPLOYMENT

The cloud-based deployment packages the entire application stack a FastAPI backend serving the model and API endpoints, a Next.js frontend providing the clinical user interface, and an nginx reverse proxy into a single Docker container hosted on Hugging Face Spaces. The architecture proposed here will reduce the dependency on the organization's server, which currently cannot be afforded by most Ugandan hospitals. The application can be accessed through any standard web browser, requiring only an internet connection and a method for capturing or uploading fundus images. Real-time endpoints for health monitoring constantly monitor the availability of the model, gate access, and inferencing time.

For facilities with dedicated computing resources, we provide Docker Compose configurations supporting GPU-accelerated inference with NVIDIA CUDA 12.1, along with optional Kubernetes manifests implementing horizontal pod autoscaling from one to eight replicas based on request load. The modular composition allows selective activation of observability tooling (OpenTelemetry, Jaeger, Prometheus), experiment tracking (MLflow), and other ancillary services through profile flags, keeping the base deployment lightweight.

B. OFFLINE-FIRST MOBILE DEPLOYMENT

Recognizing that reliable internet access cannot be assumed across much of rural Uganda, the design uses an offline-first mobile application built with Flutter for Android devices with a minimum of 4 GB RAM. The target package includes a quantized ONNX model, a FAISS vector index for offline retrieval-augmented generation, and a preloaded clinical knowledge base. On-device inference uses ONNX Runtime Mobile, and vector similarity search enables offline access to disease reference information and clinical guidelines without requiring any network connectivity.

When connectivity becomes available even intermittently over 2G or 3G networks a delta synchronization engine updates the local knowledge base by computing SHA-256 hash-based diffs against the server manifest and transmitting only changed content. The application also queues completed screening results for batch upload when connectivity permits, so no clinical data is lost due to network interruptions. A voice-first interaction mode using on-device speech recognition and text-to-speech is planned for health workers who may find spoken commands more practical than touchscreen navigation during field screening sessions.

C. UGANDA-SPECIFIC DEPLOYMENT CONSIDERATIONS

Our deployment strategy targets Uganda's existing primary healthcare infrastructure, specifically the Level III health centers and district hospitals that serve as the first point of contact for most patients in rural areas. In these hospitals, preliminary examination is done by the Village Health Teams (VHTs) and clinical officers and not ophthalmologist experts. The screening application is designed for these cadres, with an interface that minimizes clinical jargon and provides clear referral recommendations aligned with the Uganda Clinical Guidelines [26] and the Second National Health Policy [41]. Referral outputs map directly to the existing tiered referral pathway: from VHT to health center to district hospital to regional referral facility.

The separate bill of materials treats the smartphone, data plan, and basic support costs as site-level planning inputs rather than fixed national prices. Power consumption is a practical concern in areas with unreliable electricity, so the mobile application design prioritizes local inference efficiency and batched synchronization. Integration with the Uganda Health Management Information System (HMIS) through standardized DHIS2 data export formats is planned to facilitate routine reporting to District Health Officers and support population-level disease surveillance.

VII. CONCLUSION AND FUTURE WORK

This paper has presented a framework for multi-disease retinal screening designed for the realities of Ugandan healthcare delivery. By integrating a domain-specific foundation model (RETFound ViT-Large) with parameter-efficient LoRA adaptation, graph-based clinical reasoning over a 45-disease knowledge graph, and a structured orchestration pipeline, the prototype obtains a macro-averaged F1 score of 0.362 and AUC of 0.888 on the RFMiD benchmark a 12.5-fold improvement in precision over baseline approaches that collapsed to degenerate positive-for-everything predictions on RFMiD. The system provides reviewable outputs through five explainability methods, enforces input quality through a four-layer fundus image gate with a false acceptance rate below 1.5%, and maintains compliance documentation mapped against the EU AI Act and FDA SaMD frameworks. The offline-first mobile deployment strategy addresses the connectivity constraints that characterize many Ugandan health facilities outside Kampala.

Fundus Quality Gate V2

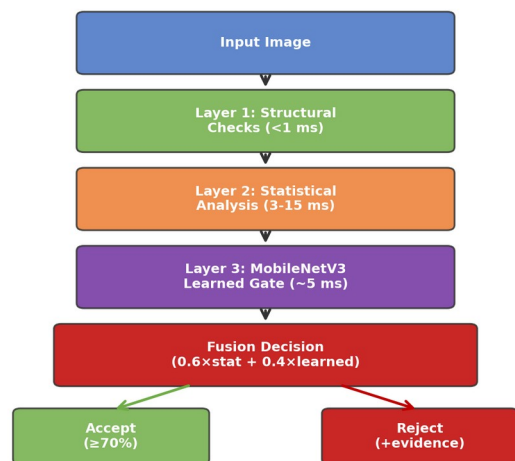


FIGURE 3. Fundus quality gate V2 architecture. Four validation layers (structural, statistical, learned MobileNetV3-Small, and fusion) filter non-fundus and low-quality images before classification, achieving a false acceptance rate below 1.5%.

Several important directions remain for future work. First and most critically, prospective clinical validation at 5 to 10 Ugandan healthcare facilities is needed, comparing model predictions against diagnoses from qualified ophthalmologists on locally collected fundus images. The RFMiD dataset on which our current results are based was not collected in East Africa, and demographic variation in retinal pigmentation, disease distribution, and imaging equipment may affect real-world performance in ways that cannot be predicted from benchmark results alone. Additionally, we recommend the use of federated learning in these hospitals by making use of the Flower or NVFlare framework for collaborative learning without the centralization of patient information. Third, active learning loops will allow ophthalmologist corrections of uncertain predictions to flow back into LoRA fine-tuning cycles, creating a continuous improvement loop that adapts the model to the local disease profile over time. Fourth, extending the system to detect systemic disease biomarkers visible in the retina such as indicators of cardiovascular risk and diabetic nephropathy could increase the public health value of routine retinal screening in Uganda. Finally, integration with Uganda's emerging electronic health record infrastructure through the DHIS2 platform will support longitudinal patient tracking and population-level epidemiological surveillance, feeding aggregate screening data back to the Ministry of Health for evidence-based resource allocation.

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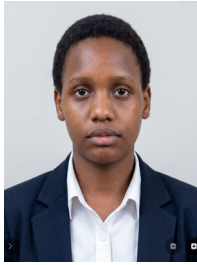
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MPAIRWE LAUBEN is enrolled at the College of Computing and Information Sciences (COCIS), Makerere University, Kampala, Uganda, where he is working towards a Bachelor of Science degree in Software Engineering. He is actively involved as the lead researcher and system architect for this project, handling model architecture, clinical knowledge graph, precision rescue approach, and MLOps pipeline. The areas of his research interest include computer vision in healthcare, application of graph neural network to clinical reasoning, design of MLOps pipelines, and deployment of AI systems in resource-limited environments.



Shadia Nankya is a student in the Bachelor of Science in Software Engineering program at COCIS, Makerere University, Kampala, Uganda. In this project, she is responsible for the following things: data pre-processing, exploratory data analysis, model evaluation, and data validation. Her research interests lie in the application of data science methods to public health research, the statistical analysis of medical images and the assessment of artificial intelligence in medicine. She was instrumental in conducting the class imbalance analysis and data validation efforts to develop the precision rescue technique.



YAPYEKO REBECCA is a prospective student pursuing a Bachelors Degree of Science in Software Engineering from the College of Computing and Information Sciences (COCIS) at Makerere University, Kampala, Uganda. YAPYEKO REBECCA supervises the deployment process, development of the mobile application, and clinical integration testing for this project. Her field of interest includes the development of mobile applications, edge computing in health care, and human computer interactions.